

Suggested Mark Scheme

- 1 (a) Describe how DNA is organised into a chromosome.

DNA double helix coiled round histone octamer forming nucleosome core;

stabilised by H1 prot forming nucleosome;

nucleosomes linked by linker DNA;

form 11nm chromatin thread / beads-on-string

coil into solenoid; form 30nm chromatin fiber;

loop on scaffold prot; form 300nm structure;

further coil in 700nm metaphase chromosome;

[4]

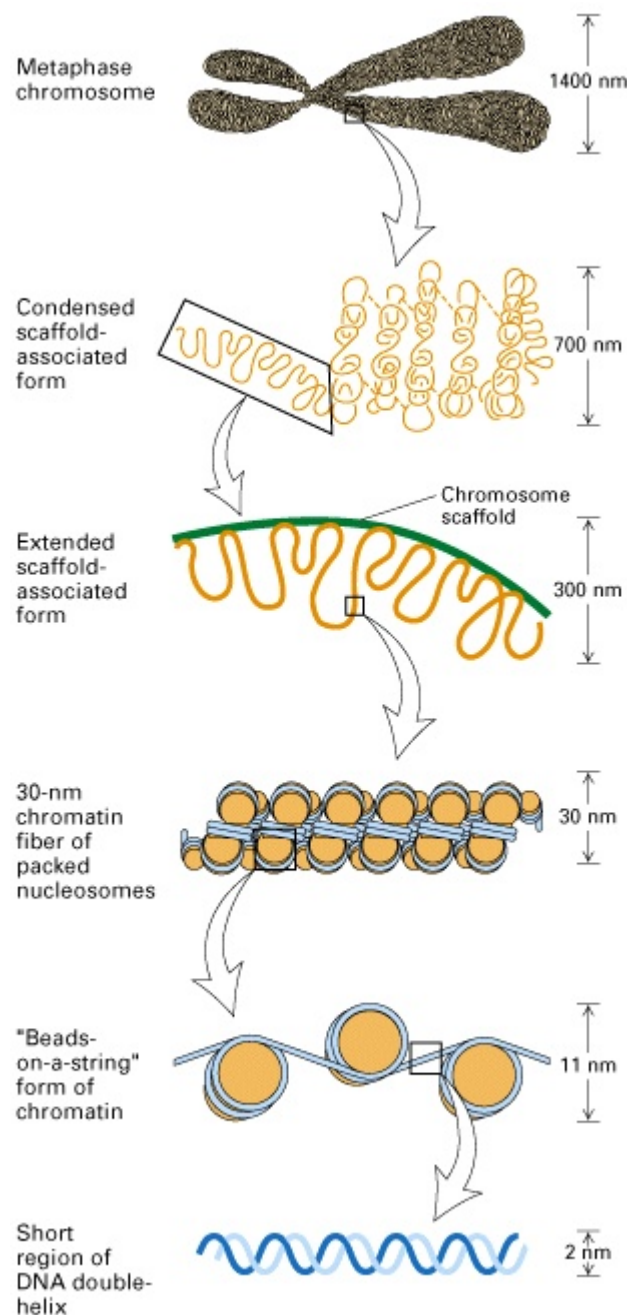


Fig. 1.1 shows part of the HIV genome.

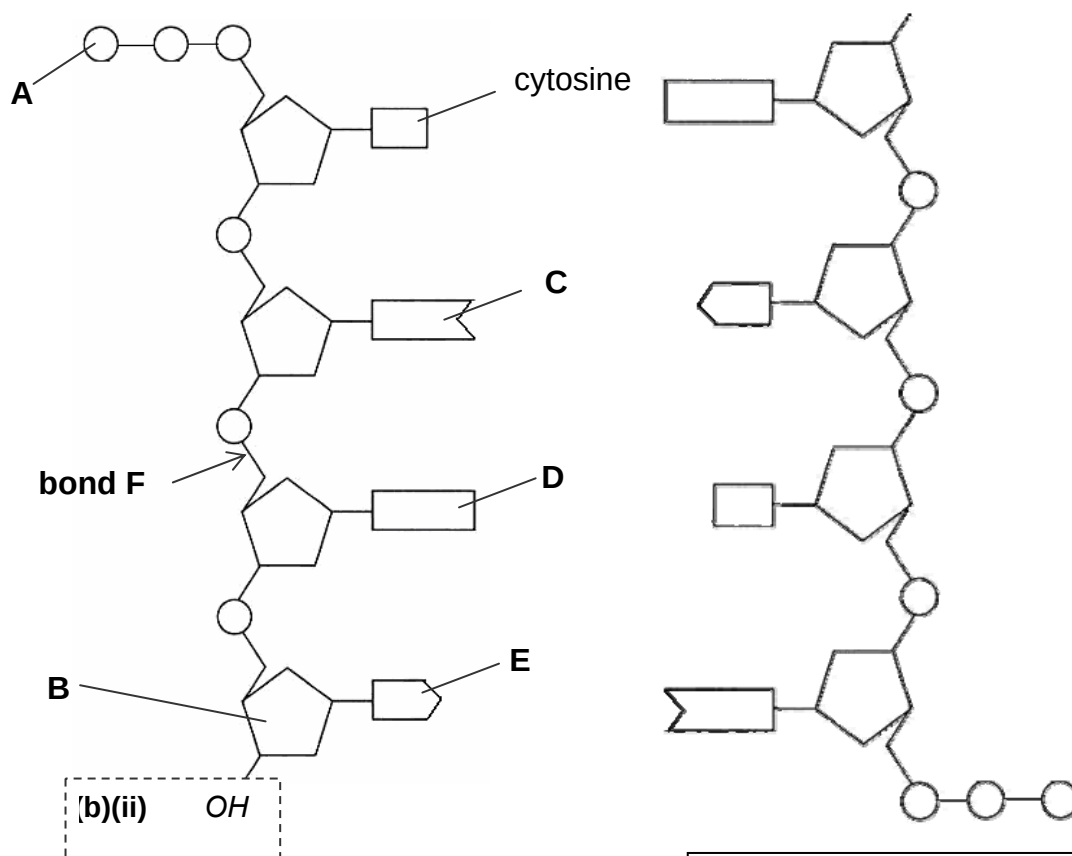


Fig. 1.1

(b)(iii) antiparallel (1m)
complementary base pairing (1m)

(b)(i) Label structures A to F.

A - phosphate

D - guanine

B - ribose

E - uracil

C - adenine

F - phosphodiester bond [3]

(ii) Write the functional group present in structure B in the box provided. [1]

(iii) Upon entering the host cell, the viral genome undergoes transcription before integration into the host genome.

On Fig. 1.1, illustrate the resultant product of this transcription from the viral genome shown. [2]

(iv) Name the enzyme used in this transcription.

reverse transcriptase; [1]

Fig. 1.2 shows the structure of the nucleotide containing the base thymine and the structure of the drug AZT. This drug is used to prevent the replication of human immunodeficiency virus (HIV) inside human cells.

Fig. 1.2

- (c) Suggest **one** way that AZT prevents replication of DNA from HIV.
acts as thymine analog, incorporated during viral genome synthesis;
but lacks 3' OH end, replication terminates [1]

[Total: 12]

2 Fig. 2 shows the life cycle of a bacteriophage.

Fig. 2

- (a) Describe events occurring in Steps 1 to 5.
- Step 1** *adsorption; tail fibres interact weakly with specific cell surface*
receptors on host cell; base plate pins form strong interaction;
contractile sheath contracts, base plate pins puncture host cell
Step 2 *membrane;*
or phage release enz to digest host cell wall / membrane;
viral genome enters via hollow tube;
viral genome directs degradation of host cell genome;
- Step 3** *biosynthesis; virus uses host ribosomes to translate viral mRNA*
to synthesize viral prot like capsid;
use host DNA pol to replicate viral genome;
- Step 4** *assembly; viral genome packaged into capsid;*
assemble tail fibres etc.
- Step 5** *lysis; cause host cell to burst by absorbing water;*
or phage release lysozyme to digest host cell wall;
viral progeny released to infect other host cells; [5]
- (b) Suggest an identity of the phage shown.
T₄ phage;; [1]
- (c) Describe the role of phages in genetic transfer between bacteria.
generalised transduction;
lytic phages erroneously package fragmented bacterial host genome into
capsid;
genes transferred are random;
specialised transduction;
erroneously excision of prophage in the lytic stage of a temperate phage;
genes adjacent to prophage integration site is packaged

transferred to 2nd bacterial host during subsequent infection;
 recombination at homologous regions may occur for both transduction;
 or integrated into 2nd host genome for specialised transduction; [4]

[Total: 10]

3 Fig. 3.1 shows the *lac* operon in *E. coli*'s genome.

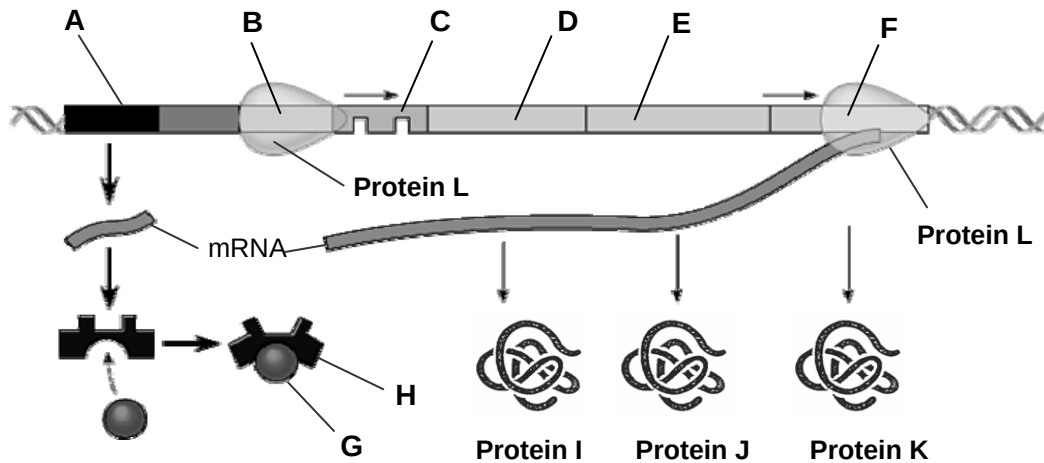


Fig. 3.1

(a)(i) Identify A – F of the DNA molecule.

A - <i>lac I</i>	D - <i>lac Z</i>	
B - <i>promoter</i>	E - <i>lac Y</i>	
C - <i>operator</i>	F - <i>lac A</i>	[3]

(ii) Identify proteins I – L.

I - <i>β-galactosidase</i>	K - <i>transacetylase</i>	
J - <i>permease</i>	L - <i>RNA polymerase</i>	[2]

(iii) Identify molecules G and H.

G - <i>allolactose</i>	H - <i>repressor</i>	[1]
® inducer / effector (too vague)		

(iv) Explain the effect of molecule G on the production of proteins I, J and K.

lac in bacterial cell is converted allolac which acts as inducer;
 binds to active repressor bound at operator site;
 cause conformational change in repressor;
 repressor-inducer cplx detaches from operator site;
 allows RNA pol to bind promoter site;

initiate transcription of lacZ, lacY and lacA;

results in the translation thus syn of prot I, J and K; [3]

Fig. 3.2 shows part of the base sequence of the mRNA resulting in the production of proteins **I**, **J** and **K**.

*

5' **AUGAUAACU.....UAAAUGGAUAAA.....UGAAUG.....UAA** 3'

Fig. 3.2

(b) With reference to Fig. 3.2,

(i) name the type of mRNA formed.

polycistronic mRNA [1]

(ii) explain how three separate proteins may be synthesised from a single mRNA.

presence of start and stop codons for each protein;; [1]

(iii) Describe the effect of a single base deletion at position indicated by * on lactose metabolism.

frameshift mutation;

results in UAA / stop codon / nonsense mutation;

truncated / no β -galactosidase but with permease & transacetylase

unable to breakdown lac into glucose + galatose; [2]

[Total: 13]

4 In the garden pea, *Pisum sativum*, the dominant alleles of two unlinked genes, **A/a** and **B/b**, are needed to make the pods tough and inedible due to lignin added to a thin cell layer. All other genotypes result in soft, edible 'sugar-snap' pea pods.

- Pods with genotypes including the dominant allele **A** have a thin layer of cells lining the pod.
- Pods with genotypes in which the recessive allele **a** is homozygous have **no** thin layer of cells lining the pod.
- Pods with genotypes including the dominant allele **B** have lignin added to the thin lining layer, when it is present, making the pods tough and inedible.
- Pods with genotypes in which the recessive allele **b** is homozygous do not have added lignin.

(a) State the phenotypes of pea pods with the following genotypes:

AAbb *soft, edible;*

aaBB *soft, edible;*

[1]

- (b) Two pea plants of genotypes **AABB** and **aabb** were interbred to give an F₁ generation and these in turn were interbred to give an F₂ generation.

Draw a genetic diagram of the cross, including gametes, showing the genotypes and phenotypes of the F₁ and F₂ generations. Give the ratio of phenotypes expected in the F₂ generation.

parental genotype	AABB	x	aabb
parental phenotype	soft, edible		soft, edible
gametes	(AB) ;		(ab) ;
F ₁ genotype	AaBb ;		
F ₁ phenotype	tough, inedible;		
F ₁ self-fertilisation	AaBb	x	AaBb
gametes	(AB) (Ab) (aB) (ab) ;		(AB) (Ab) (aB) (ab) ;
Punnett Sq;;			
F ₂ genotypic ratio	9 A ₋ B ₋ : 3 aaB ₋ : 3 A ₋ bb : 1 aabb;		
F ₂ phenotypic ratio	9 tough, inedible : 7 soft, edible;		

[5]

- (c)(i) Explain what is meant by the genes being 'unlinked'.

genes found on different chromosomes;

genes are independently segregated;

[1]

- (ii) Suggest how the phenotypic ratio of the F₂ generation might change if the two genes are linked.

does not independently segregate;

A and B alleles, a and b alleles inherited together, higher occurrence of gametes with AB and ab genotype;

F₂ with greater occurrence of parental phenotypes;

recombinant phenotypes occurs when crossing over happens;

[2]

[Total: 9]

- 5 In the kidney, glucose is freely filtered at the glomerulus and is reabsorbed in the proximal convoluted tubule. Two sodium-glucose co-transporters, SGLT1 and

SGLT2, are responsible for glucose reabsorption.

SGLT1, which is also found in the gut and other tissues, accounts for about 10% of reabsorption. SGLT2, expressed exclusively in the proximal tubule, accounts for about 90% of reabsorption.

The concentration gradient that drives the action of these transporters is driven by the Na^+/K^+ ATPase pump and glucose is transported back into the blood via the GLUT2 glucose transporter.

Fig. 5.1 summarises the absorption of glucose.

Fig. 5.1

- (a) With reference to Fig. 5.1, explain how glucose is reabsorbed from the lumen of the proximal convoluted tubule back into the blood stream.

via secondary active transport;

glucose co-transported with Na^+ via SGLT2 fr lumen into PCT cell;

glucose against conc. gradient, Na^+ down conc. gradient;

Na^+ gradient is maintained as it is actively pumped into bld stream by

Na^+/K^+ ATPase expanding energy / ATP;

glucose leaves cell into bld stream via GLUT2 via facilitated diffusion; [2]

- (b) Explain how SGLT2 is expressed exclusively in the proximal tubule.

SGLT2 gene only transcribed / active in PCT cell;;

due to interaction of activators with enhancers, TFs with promoters in PCT;;

or interaction of repressors with silencers in cells in other organs where

SGLT2 is not expressed;; [2]

Dapagliflozin, a selective SGLT2 inhibitor, is an experimental drug showing effective control in hyperglycaemia in type 2 diabetics. A placebo (PBO) was included where patients was given an inert compound which they were told was dapagliflozin. Fig. 5.2 below shows the results of a twelve week trial.

Fig. 5.2

- (c)(i) Describe the effect of dapagliflozin on blood glucose level.
dapagliflozin dose is directly proportional to the decrease in bld glucose level;

higher dose of leads to larger decrease in fasting glucose conc.;

as dapagliflozin dose increase from 2.5- 50 mg bld glucose level decrease from 12 – 25 mg/dL;

effect of dapagliflozin is difference btw PBO (-6 mg/dL) and exptal grp; [2]

- (ii) Explain how dapagliflozin aids diabetics in lowering blood glucose level.

inhibits SGLT2 in PCT preventing glucose reabsorption from lumen into bld stream;

glucose excreted in urine;; [2]

(iii) Suggest what side effects dapagliflozin may have.

excessive loss of Na⁺ in urine due to co-transport with glucose;; [1]

(iv) Suggest why a placebo was included in the study.

as negative control;; or

to offset psychological effects on measured parameter;; [1]

(d)(i) Suggest why SGLT2 instead of SGLT1 is targeted for drug design.

SGLT2 expressed exclusively in PCT;

will not affect glucose absorption in other cell types; [1]

(ii) Suggest why despite the large numbers of diabetic patients, diabetes was not a candidate for gene therapy.

may be polygenic / gene locus unknown / genetic influence uncertain;; [1]

[Total: 12]

6 Fig. 6.1 shows part of a post synaptic membrane of a cholinergic synapse.

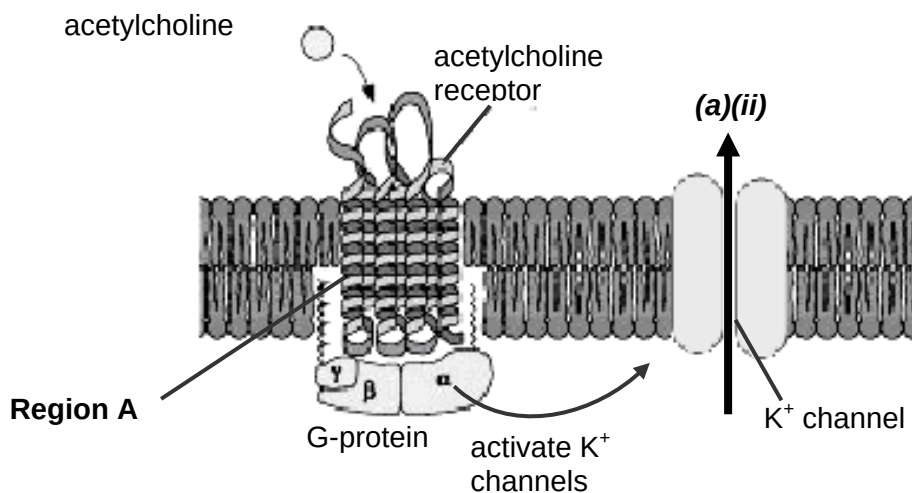


Fig. 6.1

- (a)(i) With reference to the post synaptic membrane potential, describe the events following the binding of acetylcholine to its receptor at the post synaptic membrane.

ACh binding cause conformational change in receptor;

activates α subunit of G-prot by binding GTP;

α subunit translocates to interact and activate K^+ channels;

open K^+ channels, results in efflux of K^+ fr neurone;

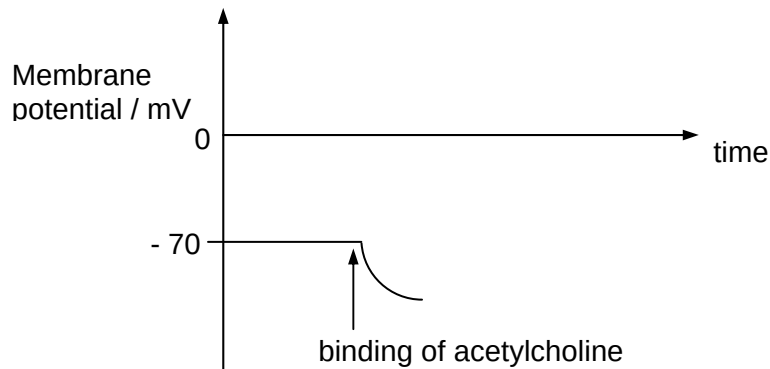
leads to more negative post synaptic mem potential;

hyperpolarisation;

cause an IPSP; [3]

- (ii) On Fig. 6.1, illustrate with an arrow, the direction of K^+ movement. [1]

- (iii) Sketch the changes in post synaptic potential in the axes provided below.[1]



- (b)(i) A mutation occurs in the α subunit of G-protein. The subunit loses its ability to hydrolyse GTP bound to it. Suggest a consequence of this mutation.

unable to convert $GTP \rightarrow GDP$;

over activation / continual activation in absence of ACh;

continued opening of K^+ channels, post synaptic mem undergoes prolonged hyperpolarisation;

nervous signal not transmitted across synapse; [2]

- (ii) Explain what kind of mutation might lead to situation described in (b)(i).

single base substitution / deletion / insertion;

leading to frameshift;

change in aa seq in α subunit of G-prot;

leading to change in 3D config of active site; [2]

(c) Describe the structure of **Region A**.

secondary prot structure, α -helix;

3.6 aa per turn;

maintained by H-bonds btw CO and NH of peptide bonds;

n+4 aa away; H-bonds parallel to long axis;

[2]

[Total: 11]

7 Table 7.1 shows the amino acid differences in the alpha haemoglobin chains between various vertebrates.

	Human	Elephant	Platypus	Ostrich	Starling	Crocodile	Lungfish	Coelacanth	Goldfish	Shark
Human		26	40	43	41	47	83	70	68	71
Elephant			45	45	48	50	84	72	63	74
Platypus				54	52	51	89	74	70	76
Ostrich					26	36	91	75	68	73
Starling						47	91	77	67	70
Crocodile							85	78	70	77
Lungfish								90	94	86
Coelacanth									83	78
Goldfish										88
Shark										

Table 7.1

Fig. 7.2 shows a family tree based on differences between the alpha haemoglobins in Table 7.1

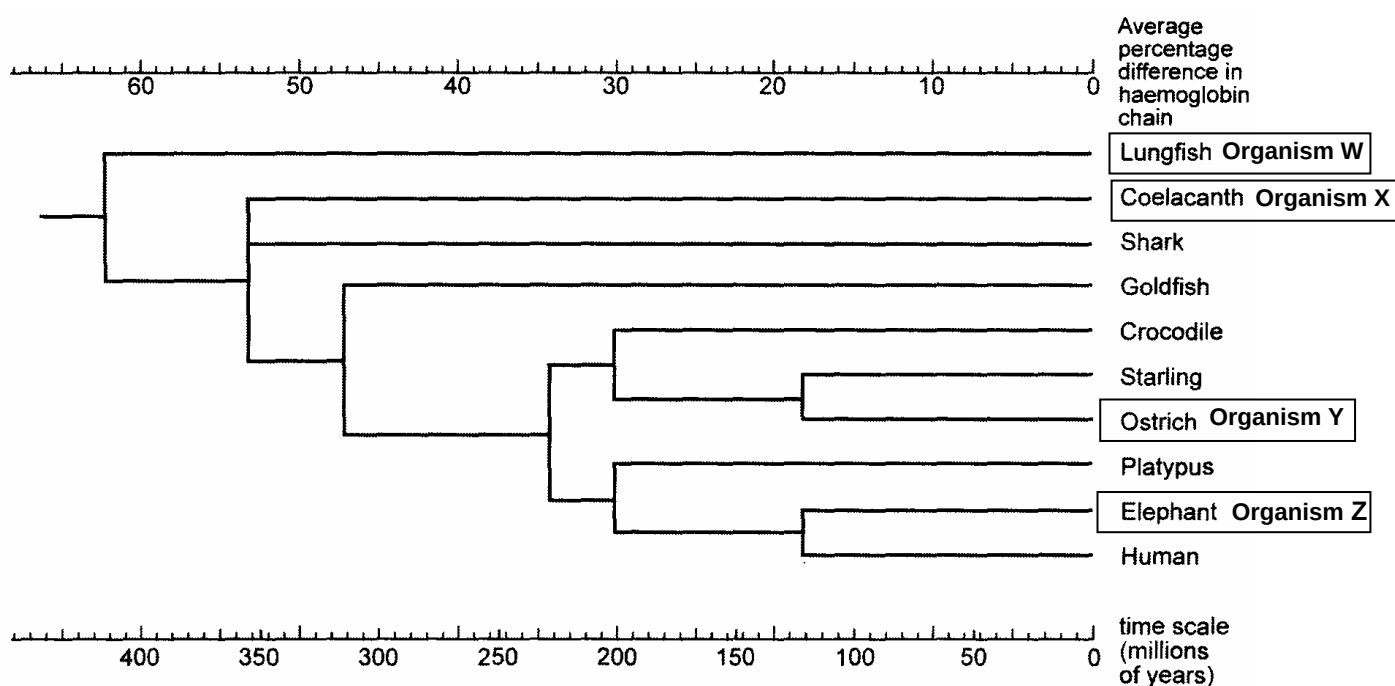


Fig. 7.2

(a)(i) Using information from Table 7.1, identify organisms **W** to **Z**.

- W *lungfish;*.....
- X *coelacanth;*.....
- Y *ostrich;*.....
- Z *elephant;*..... [2]

- (ii) Explain how differences in amino acid sequences in the alpha haemoglobin chain allow the establishment of a phylogenetic tree.

% of aa difference in Hb chain indicates relatedness;;
few difference indicates recent common ancestor / large difference
indicates early divergence;;
based on constant rate of neutral mutations acting as molecular clock;;
provides quantitative data to construct phylogenetic tree;; [3]

- (b) Explain the difference between classification and phylogeny.

classification refers to grouping organisms based on similar characteristics;
characteristics may be analogous and not homologous;
phylogeny involves grouping organisms based on evolutionary relationship;
similarity is due to inheritance from common ancestry; [2]

- (c)(i) State **two** advantages of using molecular evidence compared to homology in establishing phylogeny.

① objective;;
② quantifiable;;
③ use of phenotypically non-visible characteristics;;
④ compute small changes;; (any 2) [2]

- (ii) Suggest why homology still features prominently in evolutionary studies despite the advantages molecular evidence can confer.

① easier to use / often requires observation rather than machines;;
② DNA / prot not always available e.g. in fossils;; [2]

- (d) Explain the role of neutral mutations in evolutionary studies.

constant rate of mutation;
act as molecular clock;
genetic differences act as an indicator of time / period of divergence;
speciation event; [2]

[Total: 13]

Section B

- 8 (a) Describe the fluid mosaic model of membrane structure with reference to the functions of its components. [10]
-

1. *consists of phospholipid bilayer;*
2. *hydrophilic head facing aqueous medium inside and outside of cell;*
3. *hydrophobic tails interact to form hydrophobic core;*
4. *which restricts movement of large hydrophilic (polar, ionic) substances;*
5. *weak hydrophobic interaction btw fatty acid tails;*
6. *allows translational movement of phospholipids i.e. fluidity;*
7. *allows for cell growth, movement in unicellular organisms;*
8. *with prot asymmetrically interspersed in lipid bilayer, integral prot partially or fully embedded, peripheral prot loosely associated;*
9. *transmembranal proteins act as prot transporters / carriers, providing hydrophilic channels;*
10. *for selective entry and exit of hydrophilic substances / confer selective permeability;*
11. *act as receptor for signal molecules like hormones;*
12. *integral proteins with enzymatic activity e.g. ATP synthase*
13. *peripheral proteins for signal transduction e.g. G-prot;*
14. *with glycoproteins and glycolipids facing extracellular surface;*
15. *for cell-cell adhesion and cell-cell recognition;*
16. *cholesterol in hydrophobic region;*
17. *regulates fluidity of membrane, prevents memb from freezing at low temperature;*
18. *decreases permeability of memb by forming plug;*
19. *memb is asymmetrical in the two monolayers in terms of prot distribution and composition of phospholipids;;*
20. *labelled diagram;*
21. *AVP;*

- (b)** Using appropriate examples, explain the importance of membranes to eukaryotic cells. [4]
-

1. *allows compartmentalisation;*
2. *allows formation of localised environment optimum for specialised function;*
3. *e.g. acidic pH for lysosome / setting up of proton grad in chemiosmosis;*
4. *allows for spatial separation of reactions;*
5. *e.g. post transcriptional mRNA modifications;*
6. *containment of substance for secretion / uptake;*
7. *e.g. exocytosis of enz / endocytosis of pathogens;*
8. *regulation of entry and exit of substances;*
9. *e.g. Na^+/K^+ pump , glucose- Na^+ symport;*
10. *as attachment sites for enz for cellular rxns;*
11. *e.g. light dep rxns / ETS;*
12. *AVP;*

- (c)** Describe how enzymes like amylase are synthesised and subsequently secreted from a cell. (Details of transcription and translation are not required) [6]
-

1. *mRNA transcribed from DNA leaves nucleus through nuclear pore;*
2. *bound ribosomes on rER translates mRNA;*
3. *amylase polypeptide synthesized enters cisternal space / lumen of ER, folds into 3D config.*
4. *glycosylation of polypeptide chain;*
5. *leaves ER in transport vesicle;*
6. *vesicle fuses with cis face of GA;*
7. *amylase is sorted, packaged and modified in GA;*
8. *by adding, deleting and/or substituting sugar residues of the oligosaccharide chain and/or phosphorylation of amylase;*
9. *forming a molecular identification tag;*
10. *completed amylase leaves in secretory vesicle;*
11. *that pinches off the trans face;*
12. *moves towards cell surface membrane with help of cytoskeleton;*
13. *secreted out of cell via exocytosis;*

(@ ½ m each)

[Total: 20]

- 9 (a) Compare the structure and organization of prokaryotic and eukaryotic genomes. [8]

	<i>Prokaryote</i>	<i>Eukaryote</i>
<i>structure</i>	1. <i>single, circular chromosome (d/s DNA)</i>	1. <i>multiple linear chromosomes (d/s DNA)</i>
	2. <i>located in nucleoid region</i>	2. <i>located in nucleus (bound by double memb of nuclear env)</i>
	3. <i>no telomeres, circular chromosome, no end replication problem</i>	3. <i>presence of telomeres @ ends of chromosomes, telomere shortening, with end replication problem</i>
	4. <i>multiple extrachromosomal plasmids</i>	4. <i>cpDNA &/or mtDNA</i>
	5. <i>smaller in size (10^5 to 10^7 base pairs)</i>	5. <i>larger in size (10^7 to 10^{11} base pairs)</i>
	6. <i>DNA (-vely charged), associated with histone like proteins (+vely charged) forming loops of DNA & supercoiling</i>	6. <i>DNA (-vely charged) wound around octomer of histone proteins (+vely charged) forming nucleosome, → folding into chromatin fiber (solenoid)</i>
<i>organisation</i>	7. <i>genes (coding regions) organised into operons</i>	7. <i>genes not organised into operons, on diff chromosomes</i>
	8. <i>several structural genes under influence of single promoter</i>	8. <i>1 structural gene controlled by 1 promoter</i>
	9. <i>polycistronic → many prot encoded by 1 mRNA, many genes controlled by a single promoter</i>	9. <i>monocistronic → 1 mRNA encode only 1 prot, each gene has its own promoter and terminator</i>
	10. <i>absence of introns within genes</i>	10. <i>presence of introns within genes</i>
	11. <i>little non-coding regions, mostly coding regions, few repetitive seq</i>	11. <i>more non-coding than coding regions, many repetitive seq (e.g. centromere, telomere)</i>
	12. <i>AVP</i>	12. <i>AVP</i>

(@ ½ m per comparison)

- (b) Describe how eukaryotes regulate gene expression at chromosomal, post transcriptional and post translational level. [12]

Feature	Eukaryotic Gene Expression
Chromosome modification	<u>DNA methylation</u> 1. adding methyl grps to DNA converts C in CpG islands to 5-methyl C; 2. prevents binding of basal TFs to promoter → ↓ transcription; <u>Histone acetylation</u> 3. adding acetyl grps to histones , ↓ affinity of histone (+vely charged) for -vely charged DNA; 4. more relaxed chr conformation allowing access of TFs to promoter site → ↑ transcription; <u>Phosphorylation</u> 5. phosphorylation of lys & ser residues in histones → addition of -vely charged PO₄ grps neutralizes histones' +ve charge; 6. reduce their affinity for DNA → less coiling & easy assess by RNA pol to promoters → ↑ transcription;
Post transcriptional	<u>5' capping</u> 7. adding of methylated guanine at 5' end of 1^o mRNA forming a 5' cap; <u>3' polyadenylation</u> 8. 100 to 200 adenine (poly A) residues added to 3' end of 1^o mRNA; 9. poly(A) addition site consists of consensus seq, AAUAAA → defines position where poly(A) addition occurs; 10. protect mRNA from degradation by hydrolytic enzymes; <u>RNA splicing</u> 11. spliceosomes recognize highly conserved seq at 3' & 5' ends of introns (splice sites); 12. introns excised & exons spliced together forming mature mRNA;

	13. <i>alternative splicing</i> → include diff exons together 14. ↑ <i>single coding capacity of genome</i> → ↑ <i>diversity of prot formed</i>; <u>mRNA transport</u> 15. <i>formation of nuclear-export signal on mature mRNA with proteins</i> 16. ↑ <i>mRNA tpt thru nuclear pores</i> → ↑ <i>chances of translation</i>; <u>mRNA editing</u> 17. <i>change in mRNA base to give rise to non-functional / diff prot i.e. regulate pdtn of diff prot</i>;
Post translational	<u>Protein stability</u> 18. <i>ubiquitination of proteins</i>; 19. <i>targeted proteins are tagged</i>; 20. <i>destroyed by proteasomes</i>; <u>Protein processing/modification/transport</u> 21. <i>formation of disulphide bonds required to form active prot</i>; 22. <i>glycosylation &/or phosphorylation of prot</i>; 23. <i>proteolytic cleavage to activate protein e.g. pro-enzymes</i> 24. <i>incorrectly folded prot are retrieved back from GA into rER for degradation</i>;